RIG-ETALE IS A LOCAL PROPERTY ON ARROWS OF THE CATEGORY OF ADIC NOETHERIAN TOPOLOGICAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a preliminary base-change identification of completed naive cotangent complexes, followed by verification of the three locality conditions (1), (2), (3), with the descent step done by faithful flatness inside the delivered span

Editorial note. The label lemma-rig-etale-axioms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The proof opens with the author's sentence 'This proof is exactly the same as the proof of Lemma \ref{lemma-rig-smooth-axioms}' and then nevertheless carries out the complete concrete verification for rig-etale, including the cotangent-complex identification and all three conditions; every load-bearing step is inside the delivered span.

Editorial note. Revision r002: regenerated from the same spec after the r001 assembly crashed during packaging (stale empty job directory retained as evidence).

Allowed standard prerequisites (editorial).

- The category WAdm^{Noeth} of adic Noetherian topological rings with continuous morphisms, as used in the source section on rig-etale homomorphisms
- The completed naive cotangent complex NL with its functorial exact sequence, as developed earlier in the same chapter
- Formal Spaces: the local-property criterion for properties of arrows of WAdm^{Noeth} (the variant-Noetherian remark), and completion of etale maps staying in the category

Source reference. Original lemma 00U9: sources/stacks-chow/tags.txt, source lines 1103–1103. Prerequisite pointer invoked at proof line(s) 4848; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 4835; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0ANB: sources/stacks-chow/tags.txt, source lines 13434–13434. Prerequisite pointer invoked at proof line(s) 4815; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN9: sources/stacks-chow/tags.txt, source lines 13432–13432. Prerequisite pointer invoked at proof line(s) 4886; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0ANI: sources/stacks-chow/tags.txt, source lines 13441–13441. Prerequisite pointer invoked at proof line(s) 4791; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAF: sources/stacks-restricted/pinned-original-source.tex, source lines 560–560. Prerequisite pointer invoked at proof line(s) 4839; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AJU: sources/stacks-restricted/pinned-original-source.tex, source lines 1810–1810. Prerequisite pointer invoked at proof line(s) 4888; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00S2: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Prerequisite pointer invoked at proof line(s) 4840; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 04U5: sources/stacks-restricted/pinned-original-source.tex, source lines 2439–2439. Prerequisite pointer invoked at proof line(s) 4824; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCK: sources/stacks-restricted/pinned-original-source.tex, source lines 4253–4253. Prerequisite pointer invoked at proof line(s) 4796; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07D4: sources/stacks-restricted/pinned-original-source.tex, source lines 780–780. Prerequisite pointer invoked at proof line(s) 4851; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9G: sources/stacks-chow/tags.txt, source lines 20301–20301. Prerequisite pointer invoked at proof line(s) 4872; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The property $P(\varphi) =$"φ is rig-étale" on arrows of $WAdm^{Noeth}$ is a local property as defined in Formal Spaces, Remark 0ANI.*

Proof. This proof is exactly the same as the proof of Lemma 0GCK. Let us recall what the statement signifies. First, $WAdm^{Noeth}$ is the category whose objects are adic Noetherian topological rings and whose morphisms are continuous ring homomorphisms. Consider a commutative diagram

$$\begin{array}{ccc} B & \longrightarrow & (B')^\wedge \\ {\scriptstyle\varphi}\uparrow & & \uparrow{\scriptstyle\varphi'} \\ A & \longrightarrow & (A')^\wedge \end{array}$$

satisfying the following conditions: A and B are adic Noetherian topological rings, $A \to A'$ and $B \to B'$ are étale ring maps, $(A')^\wedge = \lim A'/I^nA'$ for some ideal of definition $I \subset A$, $(B')^\wedge = \lim B'/J^nB'$ for some ideal of definition $J \subset B$, and $\varphi : A \to B$ and $\varphi' : (A')^\wedge \to (B')^\wedge$ are continuous. Note that $(A')^\wedge$ and $(B')^\wedge$ are adic Noetherian topological rings by Formal Spaces, Lemma 0ANB. We have to show

(1) φ is rig-étale $\Rightarrow \varphi'$ is rig-étale,
(2) if $B \to B'$ faithfully flat, then φ' is rig-étale $\Rightarrow \varphi$ is rig-étale, and
(3) if $A \to B_i$ is rig-étale for $i = 1, \ldots, n$, then $A \to \prod_{i=1,\ldots,n} B_i$ is rig-étale.

The equivalent conditions of Lemma 04U5 satisfy conditions (1), (2), and (3). Thus in verifying (1), (2), and (3) for the property "rig-étale" we may already assume our ring maps satisfy the equivalent conditions of Lemma 04U5 in each case.

Pick an ideal of definition $I \subset A$. By the remarks above the topology on each ring in the diagram is the I-adic topology and B, $(A')^\wedge$, and $(B')^\wedge$ are in the category (0AL4) for (A, I). Since $A \to A'$ and $B \to B'$ are étale the complexes $NL_{A'/A}$ and $NL_{B'/B}$ are zero and hence $NL^\wedge_{(A')^\wedge/A}$ and $NL^\wedge_{(B')^\wedge/B}$ are zero by Lemma 0GAF. Applying Lemma 00S2 to $A \to (A')^\wedge \to (B')^\wedge$ we get isomorphisms

$$H^i(NL^\wedge_{(B')^\wedge/(A')^\wedge}) \to H^i(NL^\wedge_{(B')^\wedge/A})$$

Thus $NL^\wedge_{(B')^\wedge/A} \to NL_{(B')^\wedge/(A')^\wedge}$ is a quasi-isomorphism. The ring maps $B/I^nB \to B'/I^nB'$ are étale and hence are local complete intersections (Algebra, Lemma 00U9). Hence we may apply Lemmas 00S2 and 07D4 to $A \to B \to (B')^\wedge$ and we get isomorphisms

$$H^i(NL^\wedge_{B/A} \otimes_B (B')^\wedge) \to H^i(NL^\wedge_{(B')^\wedge/A})$$

We conclude that $NL^\wedge_{B/A} \otimes_B (B')^\wedge \to NL^\wedge_{(B')^\wedge/A}$ is a quasi-isomorphism. Combining these two observations we obtain that

$$NL^\wedge_{(B')^\wedge/(A')^\wedge} \cong NL^\wedge_{B/A} \otimes_B (B')^\wedge$$

in $D((B')^\wedge)$. With these preparations out of the way we can start the actual proof.

Proof of (1). Assume φ is rig-étale. Then there exists a $c \geq 0$ such that multiplication by $a \in I^c$ is zero on $NL^\wedge_{B/A}$ in $D(B)$. This property is preserved under base change by $B \to (B')^\wedge$, see More on Algebra, Lemmas 0G9G. By the isomorphism above we find that φ' is rig-étale. This proves (1).

To prove (2) assume $B \to B'$ is faithfully flat and that φ' is rig-étale. Then there exists a $c \geq 0$ such that multiplication by $a \in I^c$ is zero on $NL^\wedge_{(B')^\wedge/(A')^\wedge}$ in $D((B')^\wedge)$. By the isomorphism above we see that a^c annihilates the cohomology modules of $NL^\wedge_{B/A} \otimes_B (B')^\wedge$. The composition $B \to (B')^\wedge$ is faithfully flat by our assumption that $B \to B'$ is faithfully flat, see Formal Spaces, Lemma 0AN9. Hence the cohomology modules of $NL^\wedge_{B/A}$ are annihilated by I^c. It follows from Lemma 0AJU that φ is rig-étale. This proves (2).

To prove (3), setting $B = \prod_{i=1,\ldots,n} B_i$ we just observe that $NL^\wedge_{B/A}$ is the direct sum of the complexes $NL^\wedge_{B_i/A}$ viewed as complexes of B-modules. $\square$